

Comparative Analysis of RNN Architectures for Sentiment Classification

DATA641 HW – 3

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Dataset Summary

The experiments were conducted using the IMDb Movie Review Dataset, which contains 50,000 labeled reviews (25,00 training, 25,000 testing). Each review is labeled as *positive* or *negative*.

Preprocessing Steps:

- Text lowercased
- Punctuation and special symbols removed
- Tokenization performed using Keras tokenizer
- Top 10,000 most frequent words retained
- Sequence lengths padded/truncated to 25, 50, and 100 tokens

Model Features

Metric	Value
Embedding Dimension	100
Hidden layer size	64
Number of hidden layers	2
Dropout	0.3 to 0.5
Output layer	Sigmoid
Batch Size	32
Loss Function	Binary CrossEntropy
Optimizers	Adam, SGD, RMSProp
Activation Functions used	RELU, Tanh
Sequence Lengths	25, 50, 100
Number of Epochs trained	10
Gradient Clipping	TRUE FALSE

Dataset Statistics (after preprocessing):

Info	Insight
Average review length	~138 tokens
Vocabulary size	10,000
Training samples	25,000
Testing samples	25,000

All experiments were performed under CPU-only constraints with fixed reproducible random seeds.

Analysis of the Plots

1. Accuracy & F1 vs. Sequence Length
 - Accuracy and F1 generally increase with sequence length (25 to 100), showing that longer input contexts help the model better capture sentiment nuances.
 - Gains plateau beyond 100 tokens, suggesting diminishing returns.
2. Training Loss vs. Epochs (Best vs. Worst Models)
 - The best model (LSTM+Tanh+Adam) shows smooth and consistent loss decay.
 - The worst model (LSTM+Tanh+SGD) exhibits unstable and varying loss indicating poor convergence under SGD.

Discussion

1. **Best Configuration**
 - The LSTM architecture with Tanh activation, Adam optimizer, 100-token sequences, and gradient clipping achieved the highest F1 and Accuracy.
 - Adam yet again proved to be the best optimizer. It's adaptive learning rates enabled faster convergence compared to SGD and RMSProp under the existing CPU limitations.

2. Effect of Sequence Length

- Short sequences (25 tokens) were prone to losing context, resulting in loss of performance, indicated by the reduction in F1 by 15% on average.
- Increasing to 50 and 100 tokens improved both metrics, confirming that longer dependencies are crucial for sentiment analysis tasks.

3. Effect of Optimizer

- Adam consistently outperformed SGD, balancing stability and convergence speed.
- RMSProp yielded moderate results but higher variance across runs.

4. Effect of Gradient Clipping

- Gradient clipping improved training stability, especially in RNN and LSTM variants, which are prone to exploding gradients.
- Models without clipping occasionally performed bad.

Conclusion

Under CPU-only constraints, the **optimal configuration** is *LSTM (2 layers, hidden=64, embedding=100, Tanh activation, Adam optimizer, seq length=100, gradient clipping enabled)*

This setup achieved:

- **Accuracy:** 0.776
- **F1-score:** 0.776
- **Training time:** 27.3 seconds per epoch

The results confirm that LSTMs better handle long-term dependencies than vanilla RNNs, and Adam's optimizer provides the most robust gradient behavior for text sentiment classification tasks.